

IT342 Phase 3: Web Development Completion Report

Project Name: BidWarsOnline

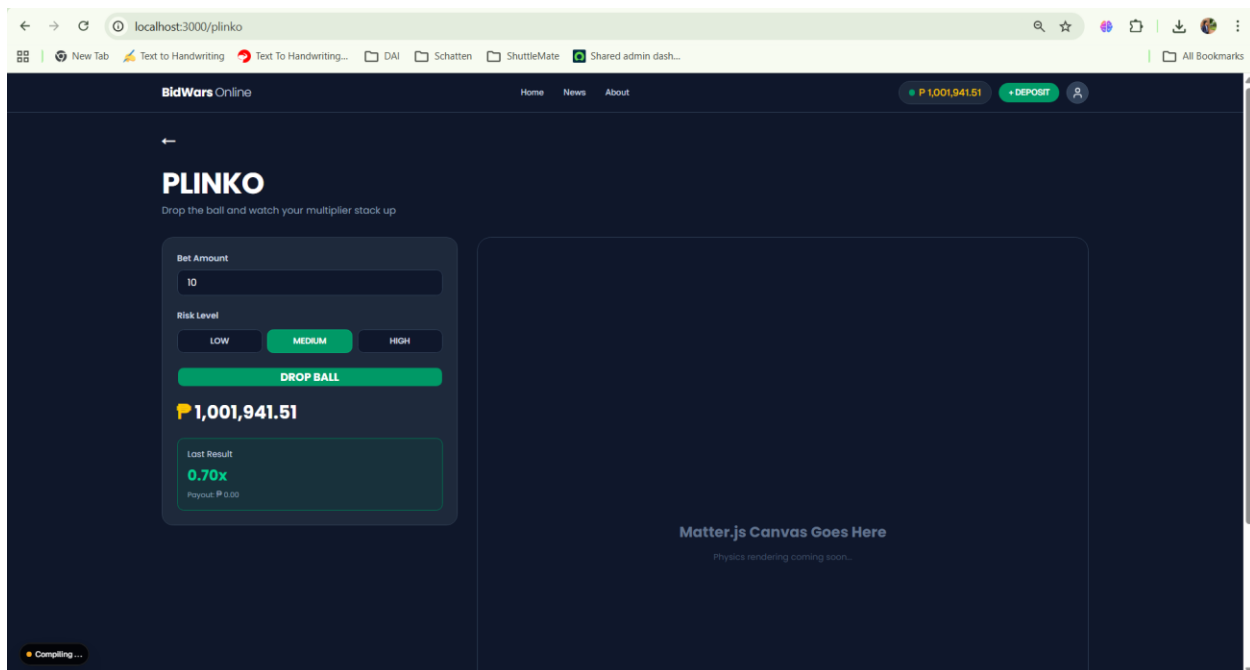
Student Name: Sanchez

1. Submission Requirements

- **GitHub Repository:** <https://github.com/Franz-Raven/IT342-Sanchez-BidWarsOnline>
- **Final Commit Message:** IT342 Phase 3 – Web Development Completed
- **Commit Hash:** <https://github.com/Franz-Raven/IT342-Sanchez-BidWarsOnline/commit/19a5ef398aca2d17a9300f616944c7cbbf661843>

2. Application Screenshots

A. Plinko Game



The user interface for playing the plinko game, matter.js UI onhold due to complexity.

Table Editor: schema public, bet_records

Filter by id, bet_amount, created_at... or ask AI

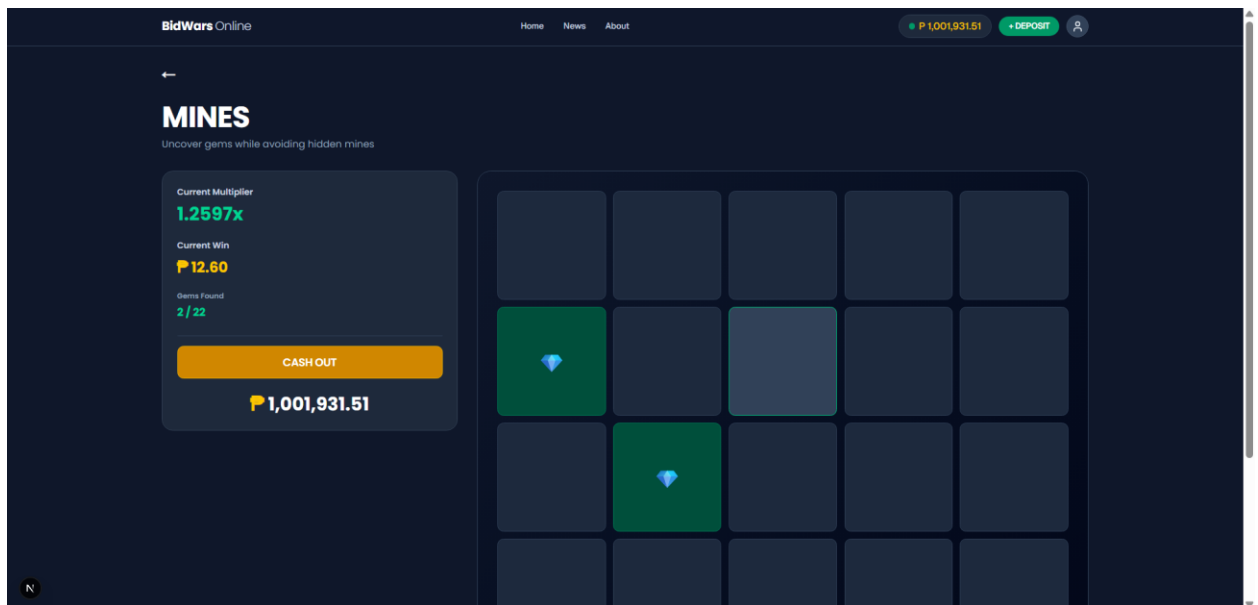
Sorted by 1 rule, RLS disabled, Index Advisor, Enable Realtime, Role postgres, Insert

id	bet_amount	created_at	game_data	game_type	multiplier	payout
106	10.0000	2026-04-12 10:25:53.290774	{"gameType":"PLINKO","risk":"MEDIUM",	PLINKO	0.7000	0.0000
105	10.0000	2026-04-12 10:12:19.785516	{"gameType":"MINES","minesCount":3,"g	MINES	1.9570	19.5700
104	10.0000	2026-04-12 10:08:00.30404	{"gameType":"HILO","result":"BUST","tim	HILO	0.0000	0.0000
103	10.0000	2026-04-12 09:41:52.50179	{"gameType":"HILO","streak":2,"timestam	HILO	4.3730	43.7300
102	10.0000	2026-04-12 09:36:01.957201	{"gameType":"HILO","result":"BUST","tim	HILO	0.0000	0.0000
101	10.0000	2026-04-12 09:34:58.98820	{"gameType":"HILO","result":"BUST","tim	HILO	0.0000	0.0000
100	10.0000	2026-04-12 09:34:23.311677	{"gameType":"HILO","streak":1,"timestam	HILO	1.2610	12.6100
99	10.0000	2026-04-12 09:34:04.72450	{"gameType":"HILO","result":"BUST","tim	HILO	0.0000	0.0000
98	10.0000	2026-04-12 09:26:29.72942	{"gameType":"HILO","result":"BUST","tim	HILO	0.0000	0.0000
97	10.0000	2026-04-12 09:19:20.245715	{"gameType":"HILO","result":"BUST","tim	HILO	0.0000	0.0000
96	10.0000	2026-04-12 09:18:43.66643	{"gameType":"HILO","result":"BUST","tim	HILO	0.0000	0.0000
95	10.0000	2026-04-12 08:59:19.489519	{"gameType":"HILO","timestamp":"2026-c	HILO	0.0000	0.0000
94	10.0000	2026-04-12 08:59:02.010182	{"gameType":"HILO","timestamp":"2026-c	HILO	0.0000	0.0000

Page 1 of 2, 100 rows, 106 records

Database record for the plinko game

B. Mines Game

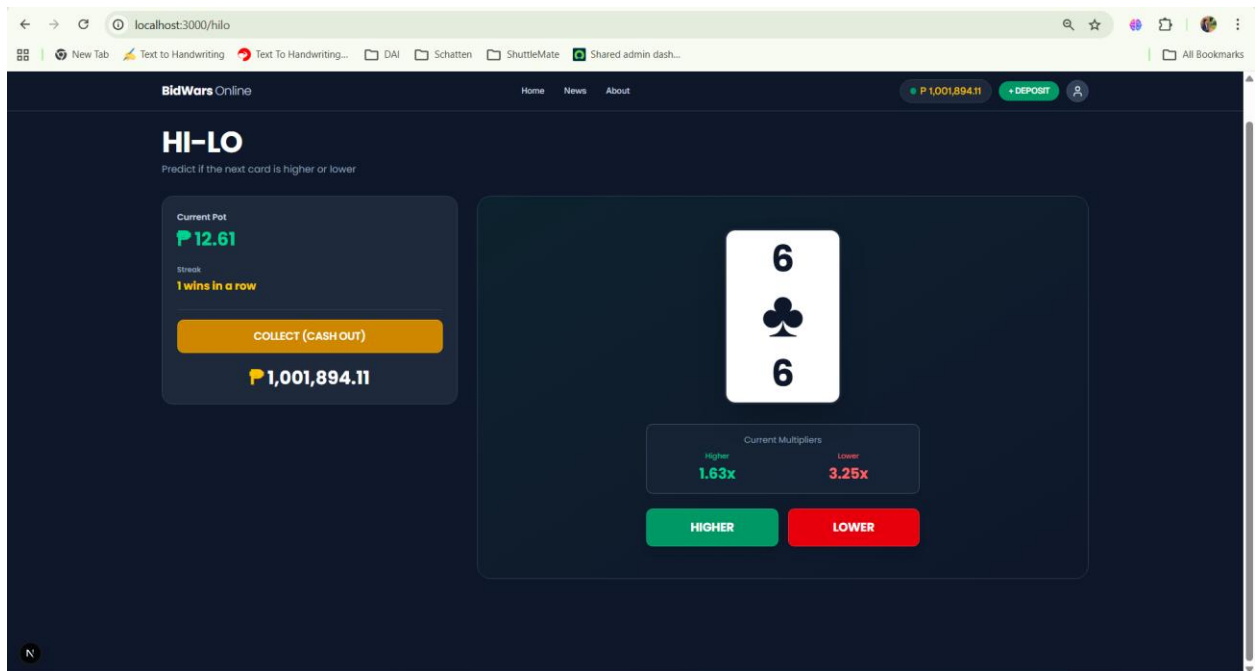


User Interface for playing the mines game

id	bet_amount	created_at	game_data	game_type	multiplier	payout
107	10.0000	2026-04-12 10:32:38.243886	{"gameType":"MINES","minesCount":3,"g	MINES	1.2597	12.6000
106	10.0000	2026-04-12 10:25:53.290774	{"gameType":"PLINKO","risk":"MEDIUM",	PLINKO	0.7000	0.0000
105	10.0000	2026-04-12 10:12:19.785516	{"gameType":"MINES","minesCount":3,"g	MINES	1.9570	19.5700
104	10.0000	2026-04-12 10:08:00.30404	{"gameType":"HILO","result":"BUST","tim	HILO	0.0000	0.0000
103	10.0000	2026-04-12 09:41:52.50179	{"gameType":"HILO","streak":2,"timestam	HILO	4.3730	43.7300
102	10.0000	2026-04-12 09:36:01.957201	{"gameType":"HILO","result":"BUST","tim	HILO	0.0000	0.0000
101	10.0000	2026-04-12 09:34:58.98820	{"gameType":"HILO","result":"BUST","tim	HILO	0.0000	0.0000
100	10.0000	2026-04-12 09:34:23.311677	{"gameType":"HILO","streak":1,"timestam	HILO	1.2610	12.6100
99	10.0000	2026-04-12 09:34:04.72450	{"gameType":"HILO","result":"BUST","tim	HILO	0.0000	0.0000
98	10.0000	2026-04-12 09:26:29.72942	{"gameType":"HILO","result":"BUST","tim	HILO	0.0000	0.0000
97	10.0000	2026-04-12 09:19:20.245715	{"gameType":"HILO","result":"BUST","tim	HILO	0.0000	0.0000
96	10.0000	2026-04-12 09:18:43.66643	{"gameType":"HILO","result":"BUST","tim	HILO	0.0000	0.0000
95	10.0000	2026-04-12 08:59:19.489519	{"gameType":"HILO","timestamp":"2026-0	HILO	0.0000	0.0000

Database record for the mines game

C. Hi-Lo game



User Interface for playing the hi-lo game

id	bet_amount	created_at	game_data	game_type	multiplier	payout
112	10.0000	2026-04-12 10:36:48.594341	{"gameType":"HILO","streak":1,"timestamp":1712944608}	HILO	1.2610	12.6100
111	10.0000	2026-04-12 10:36:21.889142	{"gameType":"HILO","result":"BUST","timestamp":1712944562}	HILO	0.0000	0.0000
110	10.0000	2026-04-12 10:36:15.221793	{"gameType":"HILO","result":"BUST","timestamp":1712944552}	HILO	0.0000	0.0000
109	10.0000	2026-04-12 10:36:08.518667	{"gameType":"HILO","result":"BUST","timestamp":1712944542}	HILO	0.0000	0.0000
108	10.0000	2026-04-12 10:36:00.998361	{"gameType":"HILO","result":"BUST","timestamp":1712944532}	HILO	0.0000	0.0000
107	10.0000	2026-04-12 10:32:38.243886	{"gameType":"MINES","minesCount":3,"timestamp":1712944298}	MINES	1.2597	12.6000
106	10.0000	2026-04-12 10:25:53.290774	{"gameType":"PLINKO","risk":"MEDIUM","timestamp":1712943953}	PLINKO	0.7000	0.0000
105	10.0000	2026-04-12 10:12:19.785516	{"gameType":"MINES","minesCount":3,"timestamp":1712943739}	MINES	1.9570	19.5700
104	10.0000	2026-04-12 10:08:00.30404	{"gameType":"HILO","result":"BUST","timestamp":1712943680}	HILO	0.0000	0.0000
103	10.0000	2026-04-12 09:41:52.50179	{"gameType":"HILO","streak":2,"timestamp":1712943112}	HILO	4.3730	43.7300
102	10.0000	2026-04-12 09:36:01.957201	{"gameType":"HILO","result":"BUST","timestamp":1712942761}	HILO	0.0000	0.0000
101	10.0000	2026-04-12 09:34:58.98820	{"gameType":"HILO","result":"BUST","timestamp":1712942718}	HILO	0.0000	0.0000
100	10.0000	2026-04-12 09:34:23.31677	{"gameType":"HILO","streak":1,"timestamp":1712942683}	HILO	1.2610	12.6100

Database record for the hi-lo game

3. Project Summary

How the Plinko Game Works

The Plinko feature is a physics-simulated drop game where users wager on where a ball will land among varying multiplier buckets. The process begins when the user selects their bet amount, risk level (Low, Medium, High), and number of rows via the Next.js interface. Upon clicking "Drop Ball," the Spring Boot backend immediately deducts the wager and calculates a Provably Fair outcome. The server transmits this outcome to the client, which then triggers the visual animation of the ball dropping into the predetermined bucket. The user's wallet is updated in real-time without requiring a page refresh.

How the Mines Game Works

The Mines feature is a grid-based risk game where users navigate a 5x5 board to uncover gems while avoiding hidden explosives. The process begins when the user sets their bet amount and selects the number of active mines via the Next.js/Kotlin interface. Upon starting the game, the Spring Boot backend securely generates a randomized grid, intentionally withholding the mine locations from the front-end to ensure provable fairness and prevent cheating. As the user clicks individual tiles, the system checks the hidden grid state. Finding a gem dynamically increases the potential payout multiplier based on combinatorial probability. Hitting a mine instantly busts the session, resulting in a loss. Users can manually trigger a "Cash Out" action after any successful click to instantly transfer their accumulated winnings to their digital wallet.

How the Hi-Lo Game Works

The Hi-Lo feature is a predictive card game built on an infinite-deck probability model. After placing an initial bet, the user is dealt a face-up card (ranks 2 through Ace). The user must predict whether

the next drawn card will be strictly "Higher" or "Lower." If correct, the potential win compounds based on dynamic odds and a scaling streak bonus (10% per consecutive win). Ties (drawing the exact same rank) result in a house win. Because the game utilizes a snowballing pot, the server maintains an active session state between guesses. The user can choose to "Collect" at any point, or the system will enforce an auto-cashout if the maximum payout ceiling is reached.

Inputs and Validations Used

Across all three games, the system captures standard financial inputs (betAmount) and game-specific configurations (e.g., riskLevel, minesCount, tileIndex, or guess). The backend rigorously validates these inputs by verifying the user has a sufficient wallet balance before any game logic executes. It also strictly enforces game boundaries (e.g., Mines count must be 1-24, tile clicks cannot be repeated, and the session status must be ACTIVE before processing a new card guess).

API Integration Used

This project utilizes a robust mix of REST APIs and WebSockets for secure, low-latency execution:

- **RESTful API:** Game initialization and stateless actions are handled via standard HTTP POST requests (e.g., /api/games/bet).
- **Real-time Capabilities:** Game execution, intermediate guessing, and balance settlements utilize STOMP WebSockets (/ws-gaming). This bidirectional communication ensures message delivery under 100ms, allowing for instant UI animations and wallet updates.

Database Tables Involved

- **Wallet:** Ensures ACID-compliant deductions of the initial bet and atomic crediting of final payouts.
- **Bet_Record:** Stores the permanent audit trail of every completed round, logging the user_id, game_type, bet_amount, multiplier, payout_amount, and a JSON snapshot of the final game_data (e.g., the final Plinko bucket, the revealed Mines grid, or the final Hi-Lo streak).
- **Session Management:** Active game states (like a multi-round Hi-Lo pot or a hidden Mines grid) are tracked in dedicated active-session entities before being finalized and logged to the permanent Bet_Record ledger.